
NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
 BY BONNIE MCBRIDE AND SANFORD GORDON
 REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

CEA analysis performed on Tue 17-Jun-2025 18:56:29

Problem Type: "Rocket" (Infinite Area Combustor)

prob case = _____5861 ro equilibrium

Pressure (1 value):

p,atm= 68

Oxidizer/Fuel Wt. ratio (17 values):

o/f = 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10

You selected the following fuels and oxidizers:

reac

fuel H2 wt%=100.0000 t,k= 298.150

oxid O2 wt%=100.0000 t,k= 298.150

You selected these options for output:

short version of output

output short

Proportions of any products will be expressed as Mole Fractions.

Heat will be expressed as siunits

output siunits

Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInputFile.cgi

IMPORTANT: The following line is the end of your CEA input file!

end

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 2.00000 %FUEL= 33.33333 R,EQ.RATIO= 3.968341 PHI,EQ.RATIO= 3.968341

	CHAMBER	THROAT
Pinf/P	1.0000	1.8215
P, BAR	68.901	37.826
T, K	2073.61	1822.50

RHO, KG/CU M	2.4165 0	1.5096 0
H, KJ/KG	0.00000	-1604.62
U, KJ/KG	-2851.29	-4110.33
G, KJ/KG	-61250.4	-55437.5
S, KJ/(KG)(K)	29.5380	29.5380
M, (1/n)	6.047	6.047
(dLV/dLP)t	-1.00007	-1.00002
(dLV/dLT)p	1.0019	1.0005
Cp, KJ/(KG)(K)	6.5196	6.2771
GAMMAS	1.2685	1.2808
SON VEL,M/SEC	1901.8	1791.4
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2547.8
CF	0.7031
Ivac, M/SEC	3190.2
Isp, M/SEC	1791.4

MOLE FRACTIONS

*H	0.00028	0.00006
*H2	0.74777	0.74795
H2O	0.25195	0.25198
*OH	0.00001	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA
CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 2.50000 %FUEL= 28.571429 R,EQ.RATIO= 3.174673 PHI,EQ.RATIO= 3.174673

	CHAMBER	THROAT
Pinf/P	1.0000	1.8059
P, BAR	68.901	38.154
T, K	2405.28	2139.58
RHO, KG/CU M	2.4287 0	1.5128 0
H, KJ/KG	-0.00000	-1582.55
U, KJ/KG	-2836.99	-4104.61
G, KJ/KG	-64836.2	-59256.6

S, KJ/(KG)(K)	26.9557	26.9557
M, (1/n)	7.049	7.054
(dLV/dLP)t	-1.00045	-1.00014
(dLV/dLT)p	1.0104	1.0037
Cp, KJ/(KG)(K)	6.1647	5.8408
GAMMAS	1.2420	1.2550
SON VEL, M/SEC	1877.1	1779.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2560.1
CF	0.6949
Ivac, M/SEC	3196.7
Isp, M/SEC	1779.1

MOLE FRACTIONS

*H	0.00163	0.00053
*H2	0.68366	0.68456
H2O	0.31456	0.31487
*OH	0.00015	0.00003

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 3.00000 %FUEL= 25.000000 R,EQ.RATIO= 2.645561 PHI,EQ.RATIO= 2.645561

	CHAMBER	THROAT
Pinf/P	1.0000	1.7909
P, BAR	68.901	38.472
T, K	2693.81	2426.12
RHO, KG/CU M	2.4729 0	1.5359 0
H, KJ/KG	-0.00000	-1540.89
U, KJ/KG	-2786.26	-4045.75
G, KJ/KG	-67098.4	-61971.3
S, KJ/(KG)(K)	24.9083	24.9083

M, (1/n)	8.039	8.053
(dLV/dLP)t	-1.00154	-1.00065

(dLV/dLT)p	1.0327	1.0150
Cp, KJ/(KG)(K)	6.1373	5.6629
GAMMAS	1.2168	1.2303
SON VEL,M/SEC	1841.3	1755.5
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2555.4
CF	0.6870
Ivac, M/SEC	3182.4
Isp, M/SEC	1755.5

MOLE FRACTIONS

*H	0.00527	0.00229
*H2	0.61791	0.62021
H2O	0.37593	0.37721
*OH	0.00089	0.00029

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA
CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 3.50000 %FUEL= 22.22222 R,EQ.RATIO= 2.267624 PHI,EQ.RATIO= 2.267624

	CHAMBER	THROAT
Pinf/P	1.0000	1.7764
P, BAR	68.901	38.788
T, K	2937.18	2678.34
RHO, KG/CU M	2.5404 0	1.5740 0
H, KJ/KG	-0.00000	-1486.58
U, KJ/KG	-2712.24	-3950.85
G, KJ/KG	-68278.8	-63748.2
S, KJ/(KG)(K)	23.2464	23.2464
M, (1/n)	9.004	9.037
(dLV/dLP)t	-1.00374	-1.00192
(dLV/dLT)p	1.0740	1.0412
Cp, KJ/(KG)(K)	6.3952	5.7630
GAMMAS	1.1945	1.2065
SON VEL,M/SEC	1799.9	1724.3

MACH NUMBER 0.000 1.000

PERFORMANCE PARAMETERS

Ae/At 1.0000
 CSTAR, M/SEC 2538.7
 CF 0.6792
 Ivac, M/SEC 3153.4
 Isp, M/SEC 1724.3

MOLE FRACTIONS

*H 0.01161 0.00627
 *H2 0.55069 0.55443
 H2O 0.43451 0.43793
 *O 0.00004 0.00001
 *OH 0.00314 0.00136
 *O2 0.00001 0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 4.00000 %FUEL= 20.000000 R,EQ.RATIO= 1.984171 PHI,EQ.RATIO= 1.984171

	CHAMBER	THROAT
Pinf/P	1.0000	1.7633
P, BAR	68.901	39.076
T, K	3137.07	2892.89
RHO, KG/CU M	2.6249 0	1.6236 0
H, KJ/KG	-0.00000	-1426.43
U, KJ/KG	-2624.87	-3833.19
G, KJ/KG	-68609.2	-64695.4
S, KJ/(KG)(K)	21.8705	21.8705
M, (1/n)	9.937	9.994
(dLV/dLP)t	-1.00719	-1.00428
(dLV/dLT)p	1.1358	1.0869
Cp, KJ/(KG)(K)	6.8709	6.1177
GAMMAS	1.1763	1.1854
SON VEL,M/SEC	1757.2	1689.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2512.5
CF	0.6722
Ivac, M/SEC	3114.0
Isp, M/SEC	1689.0

MOLE FRACTIONS

*H	0.01982	0.01256
*H2	0.48339	0.48775
H2O	0.48859	0.49538
*O	0.00018	0.00006
*OH	0.00794	0.00422
*O2	0.00007	0.00003

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 4.50000 %FUEL= 18.181818 R,EQ.RATIO= 1.763707 PHI,EQ.RATIO= 1.763707

	CHAMBER	THROAT
Pinf/P	1.0000	1.7524
P, BAR	68.901	39.317
T, K	3297.74	3069.66
RHO, KG/CU M	2.7219 0	1.6816 0
H, KJ/KG	-0.00000	-1365.46
U, KJ/KG	-2531.37	-3703.58
G, KJ/KG	-68303.8	-64945.2
S, KJ/(KG)(K)	20.7123	20.7123
M, (1/n)	10.832	10.916
(dLV/dLP)t	-1.01195	-1.00793
(dLV/dLT)p	1.2181	1.1543
Cp, KJ/(KG)(K)	7.5145	6.6867
GAMMAS	1.1623	1.1680
SON VEL,M/SEC	1715.3	1652.6
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2479.4
CF	0.6665
Ivac, M/SEC	3067.4
Isp, M/SEC	1652.6

MOLE FRACTIONS

*H	0.02833	0.02019
*H2	0.41805	0.42173
H2O	0.53673	0.54776
*O	0.00060	0.00027
*OH	0.01599	0.00992
*O2	0.00030	0.00014

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 5.00000 %FUEL= 16.666667 R,EQ.RATIO= 1.587337 PHI,EQ.RATIO= 1.587337

	CHAMBER	THROAT
Pinf/P	1.0000	1.7440
P, BAR	68.901	39.508
T, K	3423.85	3210.95
RHO, KG/CU M	2.8281 0	1.7458 0
H, KJ/KG	-0.00000	-1306.35
U, KJ/KG	-2436.27	-3569.42
G, KJ/KG	-67529.9	-64637.1
S, KJ/(KG)(K)	19.7234	19.7234
M, (1/n)	11.685	11.797
(dLV/dLP)t	-1.01802	-1.01295
(dLV/dLT)p	1.3208	1.2449
Cp, KJ/(KG)(K)	8.2931	7.4415
GAMMAS	1.1516	1.1545
SON VEL,M/SEC	1675.0	1616.4
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2441.7

CF	0.6620
Ivac, M/SEC	3016.5
Isp, M/SEC	1616.4

MOLE FRACTIONS

*H	0.03567	0.02761
H ₂ O	0.00001	0.00000
*H ₂	0.35666	0.35847
H ₂ O	0.57790	0.59355
*O	0.00149	0.00081
*OH	0.02734	0.01903
*O ₂	0.00093	0.00052

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H ₂	1.0000000	-0.000	298.150
OXIDANT	O ₂	1.0000000	-0.000	298.150

O/F= 5.50000 %FUEL= 15.384615 R,EQ.RATIO= 1.443033 PHI,EQ.RATIO= 1.443033

	CHAMBER	THROAT
Pinf/P	1.0000	1.7376
P, BAR	68.901	39.654
T, K	3519.53	3319.78
RHO, KG/CU M	2.9417 0	1.8148 0
H, KJ/KG	-0.00000	-1250.12
U, KJ/KG	-2342.22	-3435.10
G, KJ/KG	-66408.2	-63889.3
S, KJ/(KG)(K)	18.8685	18.8685
M, (1/n)	12.494	12.633
(dLV/dLP)t	-1.02523	-1.01938
(dLV/dLT)p	1.4412	1.3586
Cp, KJ/(KG)(K)	9.1571	8.3504
GAMMAS	1.1438	1.1443
SON VEL,M/SEC	1636.8	1581.2
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2401.0
CF	0.6586

Ivac, M/SEC 2963.0
Isp, M/SEC 1581.2

MOLE FRACTIONS

*H	0.04083	0.03345
H ₂ O	0.00002	0.00001
*H ₂	0.30082	0.30002
H ₂ O	0.61156	0.63165
H ₂ O ₂	0.00001	0.00000
*O	0.00304	0.00192
*OH	0.04135	0.03137
*O ₂	0.00237	0.00157

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H ₂	1.0000000	-0.000	298.150
OXIDANT	O ₂	1.0000000	-0.000	298.150

O/F= 6.00000 %FUEL= 14.285714 R,EQ.RATIO= 1.322780 PHI,EQ.RATIO= 1.322780

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.7329
P, BAR	68.901	39.760
T, K	3588.50	3399.19
RHO, KG/CU M	3.0612 0	1.8878 0
H, KJ/KG	-0.00000	-1197.26
U, KJ/KG	-2250.80	-3303.34
G, KJ/KG	-65029.9	-62796.6
S, KJ/(KG)(K)	18.1217	18.1217
M, (1/n)	13.256	13.420
(dLV/dLP) _t	-1.03306	-1.02689
(dLV/dLT) _p	1.5697	1.4893
C _p , KJ/(KG)(K)	10.0030	9.3265
GAMMAS	1.1382	1.1370
SON VEL, M/SEC	1600.6	1547.4
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2358.6
CF	0.6561

Ivac, M/SEC 2908.5
Isp, M/SEC 1547.4

MOLE FRACTIONS

*H	0.04346	0.03692
H ₂ O	0.00005	0.00003
*H ₂	0.25157	0.24803
H ₂ O	0.63764	0.66151
H ₂ O ₂	0.00001	0.00001
*O	0.00529	0.00373
*OH	0.05685	0.04594
*O ₂	0.00513	0.00384

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H ₂	1.0000000	-0.000	298.150
OXIDANT	O ₂	1.0000000	-0.000	298.150

O/F= 6.50000 %FUEL= 13.333333 R,EQ.RATIO= 1.221028 PHI,EQ.RATIO= 1.221028

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.7299
P, BAR	68.901	39.830
T, K	3634.52	3452.64
RHO, KG/CU M	3.1853 0	1.9639 0
H, KJ/KG	-0.00000	-1147.97
U, KJ/KG	-2163.07	-3176.05
G, KJ/KG	-63471.4	-61443.1
S, KJ/(KG)(K)	17.4635	17.4635
M, (1/n)	13.971	14.155
(dLV/dLP) _t	-1.04040	-1.03447
(dLV/dLT) _p	1.6887	1.6192
C _p , KJ/(KG)(K)	10.6736	10.1892
GAMMAS	1.1346	1.1321
SON VEL, M/SEC	1566.6	1515.2
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At 1.0000
CSTAR, M/SEC 2315.4
CF 0.6544

Ivac, M/SEC	2853.7
Isp, M/SEC	1515.2

MOLE FRACTIONS

*H	0.04372	0.03791
H2O	0.00008	0.00005
*H2	0.20935	0.20343
H2O	0.65656	0.68321
H2O2	0.00002	0.00001
*O	0.00812	0.00619
*OH	0.07241	0.06118
*O2	0.00973	0.00802

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 7.00000 %FUEL= 12.500000 R,EQ.RATIO= 1.133812 PHI,EQ.RATIO= 1.133812

	CHAMBER	THROAT
Pinf/P	1.0000	1.7281
P, BAR	68.901	39.872
T, K	3661.49	3484.18
RHO, KG/CU M	3.3129 0	2.0423 0
H, KJ/KG	-0.00000	-1102.29
U, KJ/KG	-2079.80	-3054.64
G, KJ/KG	-61801.2	-59910.7
S, KJ/(KG)(K)	16.8787	16.8787
M, (1/n)	14.638	14.838
(dLV/dLP)t	-1.04596	-1.04054
(dLV/dLT)p	1.7778	1.7219
Cp, KJ/(KG)(K)	11.0244	10.7216
GAMMAS	1.1324	1.1292
SON VEL,M/SEC	1534.6	1484.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2272.2
CF	0.6534

Ivac, M/SEC 2799.7
Isp, M/SEC 1484.8

MOLE FRACTIONS

*H	0.04217	0.03685
H2O	0.00013	0.00008
*H2	0.17400	0.16638
H2O	0.66909	0.69745
H2O2	0.00003	0.00002
*O	0.01128	0.00907
*OH	0.08676	0.07552
*O2	0.01655	0.01463

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 7.50000 %FUEL= 11.764706 R,EQ.RATIO= 1.058224 PHI,EQ.RATIO= 1.058224

	CHAMBER	THROAT
Pinf/P	1.0000	1.7271
P, BAR	68.901	39.893
T, K	3673.33	3498.33
RHO, KG/CU M	3.4424 0	2.1220 0
H, KJ/KG	-0.00000	-1060.10
U, KJ/KG	-2001.52	-2940.10
G, KJ/KG	-60079.7	-58277.5
S, KJ/(KG)(K)	16.3556	16.3556
M, (1/n)	15.259	15.472
(dLV/dLP)t	-1.04895	-1.04392
(dLV/dLT)p	1.8257	1.7787
Cp, KJ/(KG)(K)	11.0105	10.8149
GAMMAS	1.1312	1.1278
SON VEL,M/SEC	1504.7	1456.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2230.0
CF	0.6530

Ivac, M/SEC 2747.2
Isp, M/SEC 1456.1

MOLE FRACTIONS

*H	0.03944	0.03444
H ₂ O	0.00018	0.00012
*H ₂	0.14494	0.13635
H ₂ O	0.67625	0.70539
H ₂ O ₂	0.00004	0.00002
*O	0.01444	0.01201
*OH	0.09902	0.08777
*O ₂	0.02570	0.02389

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H ₂	1.0000000	-0.000	298.150
OXIDANT	O ₂	1.0000000	-0.000	298.150

O/F= 8.00000 %FUEL= 11.11111 R,EQ.RATIO= 0.992085 PHI,EQ.RATIO= 0.992085

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.7268
P, BAR	68.901	39.900
T, K	3673.67	3499.43
RHO, KG/CU M	3.5728 0	2.2022 0
H, KJ/KG	-0.00000	-1021.18
U, KJ/KG	-1928.47	-2832.98
G, KJ/KG	-58356.1	-56609.6
S, KJ/(KG)(K)	15.8850	15.8850
M, (1/n)	15.839	16.059
(dLV/dLP) _t	-1.04946	-1.04453
(dLV/dLT) _p	1.8345	1.7896
C _p , KJ/(KG)(K)	10.7003	10.5333
GAMMA _s	1.1308	1.1273
SON VEL, M/SEC	1476.7	1429.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2189.3
CF	0.6528

Ivac, M/SEC	2696.9
Isp, M/SEC	1429.1

MOLE FRACTIONS

*H	0.03610	0.03134
H02	0.00024	0.00017
*H2	0.12128	0.11233
H20	0.67911	0.70834
H202	0.00004	0.00003
*O	0.01736	0.01472
*OH	0.10879	0.09739
*O2	0.03707	0.03569

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 8.50000 %FUEL= 10.526316 R,EQ.RATIO= 0.933727 PHI,EQ.RATIO= 0.933727

	CHAMBER	THROAT
Pinf/P	1.0000	1.7269
P, BAR	68.901	39.899
T, K	3665.56	3491.20
RHO, KG/CU M	3.7031 0	2.2825 0
H, KJ/KG	-0.00000	-985.30
U, KJ/KG	-1860.65	-2733.38
G, KJ/KG	-56666.7	-54956.5
S, KJ/(KG)(K)	15.4592	15.4592
M, (1/n)	16.380	16.605
(dLV/dLP)t	-1.04816	-1.04317
(dLV/dLT)p	1.8152	1.7682
Cp, KJ/(KG)(K)	10.2129	10.0291
GAMMAS	1.1307	1.1273
SON VEL,M/SEC	1450.5	1403.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2150.4
CF	0.6528

Ivac, M/SEC	2649.0
Isp, M/SEC	1403.8

MOLE FRACTIONS

*H	0.03258	0.02802
H02	0.00030	0.00021
*H2	0.10211	0.09322
H2O	0.67867	0.70754
H2O2	0.00005	0.00003
*O	0.01985	0.01700
*OH	0.11607	0.10435
*O2	0.05036	0.04963

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 9.00000 %FUEL= 10.000000 R,EQ.RATIO= 0.881854 PHI,EQ.RATIO= 0.881854

	CHAMBER	THROAT
Pinf/P	1.0000	1.7271
P, BAR	68.901	39.894
T, K	3651.45	3476.49
RHO, KG/CU M	3.8323 0	2.3622 0
H, KJ/KG	0.00000	-952.24
U, KJ/KG	-1797.88	-2641.11
G, KJ/KG	-55035.4	-53350.7
S, KJ/(KG)(K)	15.0722	15.0722
M, (1/n)	16.887	17.115
(dLV/dLP)t	-1.04581	-1.04072
(dLV/dLT)p	1.7798	1.7293
Cp, KJ/(KG)(K)	9.6534	9.4386
GAMMAS	1.1309	1.1277
SON VEL,M/SEC	1425.9	1380.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2113.6
CF	0.6529

Ivac, M/SEC 2603.8
 Isp, M/SEC 1380.0

MOLE FRACTIONS

*H	0.02913	0.02478
H ₂ O	0.00036	0.00025
*H ₂	0.08656	0.07799
H ₂ O	0.67578	0.70401
H ₂ O ₂	0.00006	0.00004
*O	0.02186	0.01877
*OH	0.12108	0.10891
*O ₂	0.06518	0.06524

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

P_{in} = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H ₂	1.0000000	-0.000	298.150
OXIDANT	O ₂	1.0000000	-0.000	298.150

O/F= 9.50000 %FUEL= 9.523810 R,EQ.RATIO= 0.835440 PHI,EQ.RATIO= 0.835440

	CHAMBER	THROAT
P _{inf} /P	1.0000	1.7274
P, BAR	68.901	39.886
T, K	3633.14	3457.39
RHO, KG/CU M	3.9602 0	2.4410 0
H, KJ/KG	0.00000	-921.75
U, KJ/KG	-1739.86	-2555.77
G, KJ/KG	-53475.9	-51810.8
S, KJ/(KG)(K)	14.7189	14.7189
M, (1/n)	17.362	17.592
(dLV/dLP) _t	-1.04299	-1.03784
(dLV/dLT) _p	1.7373	1.6836
C _p , KJ/(KG)(K)	9.0889	8.8447
GAMMA _s	1.1313	1.1282
SON VEL, M/SEC	1402.9	1357.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2078.9
CF	0.6531

Ivac, M/SEC	2561.2
Isp, M/SEC	1357.8

MOLE FRACTIONS

*H	0.02590	0.02178
H02	0.00041	0.00029
*H2	0.07388	0.06576
H2O	0.67109	0.69857
H2O2	0.00006	0.00004
*O	0.02338	0.02004
*OH	0.12414	0.11146
*O2	0.08114	0.08205

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 10.00000 %FUEL= 9.090909 R,EQ.RATIO= 0.793668 PHI,EQ.RATIO= 0.793668

	CHAMBER	THROAT
Pinf/P	1.0000	1.7278
P, BAR	68.901	39.877
T, K	3611.97	3435.35
RHO, KG/CU M	4.0861 0	2.5187 0
H, KJ/KG	0.00000	-893.60
U, KJ/KG	-1686.21	-2476.84
G, KJ/KG	-51994.5	-50345.7
S, KJ/(KG)(K)	14.3951	14.3951
M, (1/n)	17.810	18.041
(dLV/dLP)t	-1.04005	-1.03491
(dLV/dLT)p	1.6930	1.6370
Cp, KJ/(KG)(K)	8.5537	8.2864
GAMMAS	1.1317	1.1288
SON VEL,M/SEC	1381.4	1336.9
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2046.3
CF	0.6533

Ivac, M/SEC	2521.2
Isp, M/SEC	1336.9

MOLE FRACTIONS

*H	0.02296	0.01907
H02	0.00046	0.00033
*H2	0.06348	0.05587
H20	0.66512	0.69179
H202	0.00007	0.00004
*O	0.02443	0.02086
*OH	0.12560	0.11239
*O2	0.09789	0.09966

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS